

We resume the very figure of the last lemma. From the point A springs the arc ACB , fixed in position, with smooth continuous curvature; across it runs the chord AB , and grazing the curve at A stretches the tangent AD . As the point B slides toward A , all three of these — arc, chord, and tangent — shrink together toward nothing, and the question is: in what *proportion* do they vanish? Newton's claim is bold and exact: their ultimate ratio, any one to any other, is the ratio of equality — $1 : 1 : 1$. The trouble is that three quantities all collapsing to zero cannot be compared directly, so we need a clever device to hold their proportions still while we examine them.